



Rules

- There is no time limit.
- You may use the internet.
- It is prohibited to have any form of communication with another person.

Question 1

Create an object oriented solution for the following problem:

- Create an API for a Bicycle, to make it cycle. Additionally it must be possible to increase and decrease the bicycle's gear.
- Use the Bicycle.php class given in Appendix A. Fill in its methods.
- Feel free to create additional code: classes, properties, constants, methods and constructors if you feel that is necessary.
- Can you provide automated tests for your code?

Additional information:

- The bicycle's moving distance after one wheel rotation is denoted as "*[Rear wheel circumference] * [Gear ratio]*".
- The gear ratio is denoted as "*[Number of chain ring teeth] / [Number of sprocket teeth]*".
 - The chain ring is the cog wheel in the middle of the bike between the pedals.
 - The sprocket wheels are attached to the middle of the rear wheel. When a bicycle switches gears, the chain moves to a different sprocket wheel.
- Take into account that it should be possible to create bicycles with different gear equipment:
 - Cheap gear, with
 - 7 gear levels
 - Number of sprocket teeth: $30 - 2 * \$gear_level$
 - Expensive gear, with
 - 30 gear levels
 - Number of sprocket teeth: $46 - floor(1.2 * \$gear_level)$
- Take into account that it should be possible to create bicycles with several different wheel sizes. Examples of such wheel sizes are:
 - 20 inch
 - 24 inch
 - 28 inch

Question 2a

Design a database structure which holds the following information. Please make sure to take into account the rules of database normalization.

- Persons must be stored.
- Cars must be stored.
- A person has a first name and a last name. Together these identify the person.
- A car has a license plate.
- Some persons own cars.
- Persons may even own multiple cars.
- There are different types of cars: sedans and trucks.

Question 2b

Create another database structure, that continues on the structure provided by question 2a. The following information must be added:

- A car has a brand. Examples: 'Toyota', 'Opel'.
- A car has a brand type. Examples: 'Corolla', 'Corsa'.
- Truck brand types must store a loading capacity.
- Sedan brand types must store the passenger capacity.

Appendix A

```
<?php
class Bicycle {

    /**
     * Increases the gear level with 1.
     */
    public function increaseGear(): void {

    }

    /**
     * Decreases the gear level with 1.
     */
    public function decreaseGear(): void {

    }

    /**
     * Returns the length traveled in meters
     */
    public function cycle(int $number_of_rotations): float {

    }
}
```